

InfraRisk Monitor

Comprehensive Project Documentation & Technical Architecture Specification

1. Executive Summary & Problem Context

Centrally-sponsored mega infrastructure projects across India (Roads & Highways, Railways, Power, Urban Infrastructure, and Irrigation) consistently suffer from severe schedule slippages and cumulative cost escalations. According to CAG and Ministry of Statistics and Programme Implementation (MoSPI / PAIMANA) audits, over 40% of major projects exceed sanctioned budgets, primarily due to latent physical construction delays that remain undetected during initial execution stages.

InfraRisk Monitor is an AI-powered predictive governance platform designed to bridge this critical information gap. By evaluating non-linear interactions between physical construction progress, fund drawdowns, time elapsed, and historical sector patterns, InfraRisk Monitor flags impending cost and milestone overruns 6 to 12 months in advance.

2. System Architecture & Technical Specifications

The system is built on a modern, decoupled client-server architecture engineered for high throughput, strict type safety, and real-time interactive analytics.

Component	Technology	Role / Key Capabilities
Frontend Framework	Next.js 14 (App Router)	Server and Client React components, optimized production builds
UI & Styling	Tailwind CSS + Lucide	Clean sans-serif design system, responsive grids, color risk badges
Data Visualizations	Recharts	Step-charts, dual-axis progression plots, and horizontal bar charts
Backend Framework	FastAPI (Python 3.12)	Asynchronous REST microservice with Pydantic v2 schemas
Machine Learning	scikit-learn 1.4.2	Gradient Boosting Classifier (87.5% acc) & Regressor (MAE 4.60%)
Data Processing	Pandas 2.2.2 & NumPy	In-memory dataset filtering, CSV staging, and dynamic aggregation
Conversational AI	Hybrid RAG Query Engine	In-memory semantic NLP parser + Google Gemini LLM API connector

3. Machine Learning Methodology & Benchmarks

InfraRisk Monitor incorporates a dual-model machine learning architecture calibrated against 800 synthetic infrastructure packages generated with seed 42, accurately calibrated to historical PAIMANA project distributions.

Model Evaluation Benchmark:

Model Name	Model Type	Error (MAE)	Accuracy	Performance vs Baseline
Gradient Boosting Regressor	AI Ensemble (100 trees)	4.60%	—	+28.6% Improvement
Gradient Boosting Classifier	AI Risk Categorizer	—	87.5%	High / Medium / Low
Linear Regression	Parametric Baseline	4.79%	—	Underfits late acceleration
Statistical Moving Average	Historical Mean	6.44%	—	Unresponsive to early lags

4. Feature-by-Feature Technical Specifications

Detailed technical specifications and user-facing workflows for all 10 platform modules:

Feature 1: Functional CSV Ingestion & Pre-Commit Staging

- File Parsing & Header Mapping: Automatically maps varying CSV column aliases ('cost', 'agency', 'elapsed_time') to canonical database schemas.
- Ratio-to-Percentage Normalization: Converts ratios (0–2 scale) to standard percentages (0–100%).
- Pre-Commit Analytics Preview: Computes progress lag, risk distribution counts (High, Medium, Low), and average cost overrun before saving.
- Top At-Risk Project Table: Visual preview of the most vulnerable rows in the uploaded file.
- One-Click Database Ingestion: 'Add these projects to dashboard' merges parsed records into the central persistent database.

Feature 2: Revised Capital Allocation & Money Tracking

- Hierarchical Financial Structure: Stores original_sanctioned_cost alongside an array of sequential revisions {date, revised_cost, reason}.
- Escalation Metrics: Displays original sanctioned cost, latest approved budget, and cumulative percentage escalation.
- Trajectory Step-Chart: Recharts AreaChart visualizes capital step-ups over the project lifecycle.
- Revision Audit History: Logs official statutory justifications (e.g. land acquisition compensation, material price indexation, scope changes).

Feature 3: Monthly Progress & Budget Comparison View

- Milestone Progression Time-Series: Simulates 8 to 12 monthly snapshots per project {month, revised_cost, physical_progress_pct, fund_utilization_pct}.
- Dual-Axis Comparative Chart: Simultaneously plots physical progress % and fund utilization % against approved budget (■ Cr).
- Disparity Diagnostics: Visually exposes early phases where capital disbursements outpace ground asset formation.

Feature 4: Smart Timeline & Gantt View with Schedule Slip Overlay

- Milestone Tracking: Displays Sanctioned Start Date, Today's Date Marker, Target Commissioning Date, and AI-Predicted Completion.
- Prominent Red Schedule Slip Overlay: Highlights the exact temporal gap (+X months) between contractual delivery and AI pace projections.
- Visual Schedule Slip Alerts: Provides an immediate visual indicator of delays, far surpassing static Gantt charts.

Feature 5: Configurable Early-Warning Rules Engine

- Elimination of Hardcoded Rules: Projects are dynamically evaluated in real-time against active threshold rules in /api/alerts/.
- Active Core Rules: Severe Progress Lag (> 20%), Budget Overrun (> 25%), Expenditure Disparity (> 15%), and Timeline Slippage (> 8 mo).

- Admin 'Manage Rules' Panel: Authorized administrators can toggle rules on/off and fine-tune numerical thresholds dynamically.

Feature 6: Role-Based Access Control (Admin vs Viewer)

- Client-Side State Security: Persistent RoleContext (Admin | Viewer) stored in localStorage.
- Top-Navbar Toggle Pill: One-click interactive switcher to demonstrate access rights to jury members.
- Viewer Protection: Non-admins see read-only notifications on /admin and cannot modify rule thresholds.

Feature 7: Automatic Project Audit Report Generator (PDF)

- One-Click Printable Summary: Project Detail includes a 'Generate Project Report (PDF)' button.
- Print-Optimized Layout: Clean executive report formatting hiding navigation bars and optimizing diagnostic cards for physical/digital PDF print.

Feature 8: What-If Risk Simulator

- Interactive Sensitivity Testing: Allows policy-makers to adjust Physical Progress %, Fund Utilization %, and Time Elapsed % via real-time sliders.
- Live ML Model Evaluation: Instant asynchronous calls to /api/projects/simulate re-evaluating risk labels and cost overruns.
- Delta & Risk Driver Tracking: Shows exact overrun reduction/increase against baseline alongside explainable AI feature weights.

Feature 9: Agency Accountability Leaderboard

- Cross-Agency Benchmarking: Ranks NHAI, Indian Railways, NTPC, Municipal Corporations, State PWD, and CPWD.
- Accountability Grading: Automatic assignment of performance grades from Grade A (Low Overrun) to Grade D (Critical Overruns).
- Visual Overrun Comparison: Recharts bar chart ranking agencies by average cost overrun % and schedule delay.

Feature 10: Data Quality & Reporting Compliance Score

- Data Freshness Tracking: Tracks last_updated_date and flags projects inactive for > 60 days.
- Dashboard Compliance Stat Card: Summary card displaying 'X projects with overdue reporting'.
- Project Explorer Filtering: Dropdown filter for 'Overdue Reporting' with prominent amber warning badges.

5. AI Assistant & Conversational Intelligence

InfraRisk Monitor features an advanced Natural Language Processing assistant accessible via the dedicated /chat page and a global floating widget on every screen.

Capabilities:

- Agency Performance Benchmarking: 'Which agency is performing the best?' calculates real-time averages and builds ranked tables.
- Head-to-Head Comparisons: 'Compare NHAI vs Indian Railways' computes comparative metric matrices across both agencies.
- Outlier Discovery: Inquiries regarding highest delay, highest cost overrun, or largest budget pull exact project records.
- Project Diagnostics: Asking about any ID (e.g. 'PRJ0015') returns full risk drivers and ground progress lag.
- Financial Aggregation: Queries regarding total portfolio sanctioned budget (■ Crores) and sector distributions.
- Hybrid Architecture: Deep in-memory Pandas analytical query engine + Google Gemini LLM API connector.

6. Strategic National Impact & V2 Roadmap

- ✓ Capital Conservation: Early intervention on just 5% of at-risk mega projects saves thousands of crores for the national exchequer.
- ✓ Predictive Rather than Autopsy: Shifts governance from historical post-mortem audits to proactive real-time mitigation.
- ✓ Inter-Agency Synergy: Transparent accountability benchmarking eliminates inter-departmental finger-pointing.
- ✓ V2 Roadmap — Satellite Earth Observation: Verification of ground physical milestones using Sentinel-2 orbital imagery.
- ✓ V2 Roadmap — Weather & Monsoon AI: Live rain forecast APIs predicting monsoon-driven construction halts.
- ✓ V2 Roadmap — PFMS / MoSPI Integration: Real-time API integration with the Public Financial Management System.